Exploring the human diseasome: the human disease network

Kwang-Il Goh and In-Geol Choi

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Abstract

Advances in genome-scale molecular biology and molecular genetics have greatly elevated our knowledge on the basic components of human biology and diseases. At the same time, the importance of cellular networks between those biological components is increasingly appreciated. Built upon these recent technological and conceptual advances, a new discipline called the network medicine, an approach to understand human diseases from a network point-of-view, is about to emerge. In this review article, we will survey some recent endeavours along this direction, centred on the concept and applications of the human diseasome and the human disease network. Questions, and partial answers thereof, such as how the connectivity between molecular parts translates into the relationships between the related disorders on a global scale and how central the disease-causing genetic components are in the cellular network, will be discussed. The use of the diseasome in combination with various interactome networks and other disease-related factors is also reviewed.

Keywords: *diseasome; human disease network; interactome; cellular network; network medicine*

INTRODUCTION

Knowledge organization often begins with categorization. The classification of human disease has long been based more often on empirical phenotypic features including symptoms, anatomy or physiology of the disease, than on its molecular etiology [1]. Advances in genetics and molecular biology allowed describing human diseases in terms of molecular and genotypic features [2, 3]. The categorization is by nature a holistic process, requiring a global picture of organization as well as mechanistic details of individual components. In human disease, the combinatorial analysis of genotypic–phenotypic relationships at the global level can provide such a comprehensive view. This global view may in turn yield novel insights into the cause and effect of diseases.

In company with innovative technologies including next-generation sequencing and genome-wide association study, an explosive number of studies are ongoing to link genomic loci with specific disease phenotypes and thus discover 'disease genes' or genetic traits associated with human diseases [4, 5]. Those genotype–phenotype associations related with human diseases are being accumulated in such databases as the Online Mendelian Inheritance in Man (OMIM; http://www.ncbi.nlm.nih.gov/omim) [6] and the genetic association database (GAD; http://geneticassociationdb.nih.gov/) [7], covering over 12 000 genes. These gene-disease association data should encode intrinsic features of diseases and cognate genes. Based on genetic foundation, human diseases can be divided into two

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In-Geol Choi is a professor leading the Computational and Synthetic Biology Laboratory in the Division of Biotechnology at Korea University, Korea. He is interested in mapping the human disease universe where all genic and phenetic relationships among diseases are revealed. From the global view of human diseasome, he attempts to survey complex diseases and comorbidity, and categorize human disorders in an evolutionary perspective.

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categories: monogenic and polygenic diseases. Monogenic disease is caused by mutations in a single gene, where phenotypic effect is usually obvious as is in a Mendelian disease. In contrast, polygenic disease originates from mutations in multiple genes, where phenotypic expression is often cumulative or cooperative. Many complex diseases (e.g. diabetes, cancer and heart disease) are polygenic diseases. On the other hand, different mutations in a single gene may cause multiple disease phenotypes (e.g. various cancer cases of disordered *TP53* [8]) while a single disease phenotype can be caused by various genes (e.g. Zellweger syndrome caused by any of 11 genes related to biogenesis of peroxisome [9]). Due to the intrinsic complexity of genotype–phenotype associations, the cause and effect of disease become more ambiguous and difficult to elucidate underlying mechanisms. As such, organizing individual disease-gene association data is becoming increasingly complicated and the necessity of a global view of relationships among diseases and genetic components has become inevitable.

On this new wave of disease genetics, the complex network theory, a branch of mathematics and theoretical physics, plays an instrumental role by providing conceptual insights as well as offering visual and computational methodology [10–13]. It is thus natural to place the human disease studies onto network perspective, which fully embraces the complexity and connectivity of biological processes. In this regard, a conceptual platform to project such associations in its entirety, called the human diseasome, has recently been introduced, which links all disease phenotypic features (human disease phenome) to all known disease genes (human disease genome) [14]. The main goal of this review article is to provide the readers with an overview and motivate them to engage in this exciting new field. This review is by no means meant to be complete or exhaustive. An interested reader may also be referred to some recent review articles on related topics [15–20].

THE HUMAN DISEASOME

The diseasome can best be represented by a bipartite graph consisting of two disjoint sets of nodes. The first set is the disease nodes and the other is the disease gene nodes. A disorder and a gene are then connected by a link if mutations in that gene are implicated in that disorder. A bipartite graph is a graph in which the links always connect the two nodes, each from two disjoint sets of nodes, as is the case for the diseasome. From the bipartite diseasome (Figure 1, middle), one can construct the human disease network, the network of human diseases connected by sharing common genetic component (Figure 1, left). One can also construct the human disease gene network, the network of human genes, connected by implicating common human disorders (Figure 1, right). The first version of the diseasome was created based on the list of human disorders, disease genes and associations between them obtained from the OMIM database as of December 2005 [14]. OMIM initially focused on monogenic disorders and has only recently expanded to include complex traits and the genes mutations of which confer susceptibility for these common disorders, so the current disorder disease-gene associations are biased towards those transmitted in a Mendelian manner. Other databases, such as GAD [7], provide a complementary aspect to the OMIM data [21].

The obtained diseasome network revealed several key characteristics in global organization of the human diseasome (Figure 2). First, human diseases are found to be highly connected genetically, displaying many connections between both individual disorders and disorder classes. Among 1286 disorders in the dataset, 869 (>67%) have at least one link to other disorders and 516 (40%) disorders form a single, large cluster. This suggests indeed there may be a widespread genetic relatedness across many domains of human disorders, transcending traditional disease categorization. Second, despite intensive networking, connections between disorders are not completely random. Rather, disorders tend to form clusters by similar pathophysiology. The most outstanding example is the large cluster of cancer and cancer-related disorders, which is tightly interconnected through several common oncogenes and/or tumor suppressor genes (*TP53*, *KRAS*, *ERBB2*, *NF1*, etc.). These disorder clusters with relatively homogeneous pathophysiological characteristics are connected on a larger scale to form the largest disease cluster spanning 40% of known disorders. To achieve this global connectivity, complex diseases such as diabetes and obesity play the role of 'connectors,' bridging disorders in different classes. Third, there is high heterogeneity in the level of both disorders and genes, suggesting a broad spectrum in the degree of genetic heterogeneity associated with human disease processes. For example, while most disorders are

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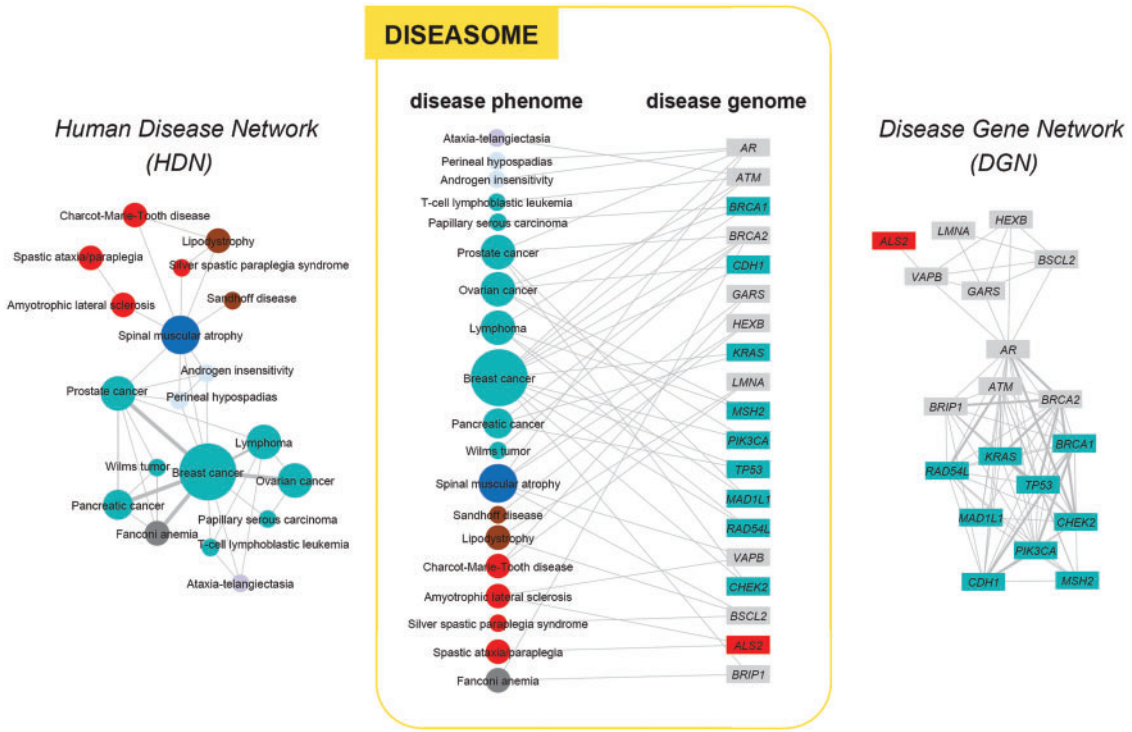


Figure 1: Schematic illustration of the human diseasome. Disease phenome and disease genome form the two node sets of bipartite network in which a disorder node and a gene node are connected if the gene is implicated in the disorder. From the bipartite diseasome (middle), one can project it to form the human disease network (left) or the human disease gene network (right). Adapted from [14]. Copyright (2007) National Academy of Sciences, USA.

associated with only a few disease genes, some disorders such as deafness, leukaemia and colon cancer are associated with more than 30 disease genes, resulting in a highly skewed distribution of number of associated disease genes per disorder. Likewise, whereas most genes are involved in only a few disorders, several disease genes (e.g. *TP53*, *PAX6*) are involved in as many as 10 disorders, being major hub genes in the network. Similar features were observed also in other diseasome-related networks [22–31]. The network-based approach is increasingly applied to particular disease classes such as autoimmune, neurological or cardiovascular diseases [32–34].

CELLULAR NETWORKS AND THE DISEASE NETWORK

Modular organization has been formulated in several layers of the molecular interactome network [10–12]. Given that such internal modular organization of molecular components should, at least to some extent, translate to external phenotypic outcomes, one can naturally expect a modular nature also of human diseases [35, 36]. Supporting evidence to this reasoning comes from several genetic disorders known to arise from mutations in genes participating in the same cellular pathway, molecular complex or functional module. For example, Fanconi anaemia arises from mutations in a set of genes encoding proteins involved in DNA repair, many of them forming a single heteromeric complex [37]. However, the question of how generically human disorders and disorder classes would correspond to discernable functional modules in the global interactome network is not fully answered yet, but some indirect evidence exists [14]. For example, it was shown that the genes associated with the same disease tend to encode proteins that are more likely to interact with one another than with other proteins in the proteome. Also, those genes have a higher tendency to share common Gene Ontology terms. Moreover, they tend to be expressed in the same set of human tissues, as well as showing higher correlation in expression profile across tissues, than expected by chance alone. Built upon these ideas and observations, the so-called disease module

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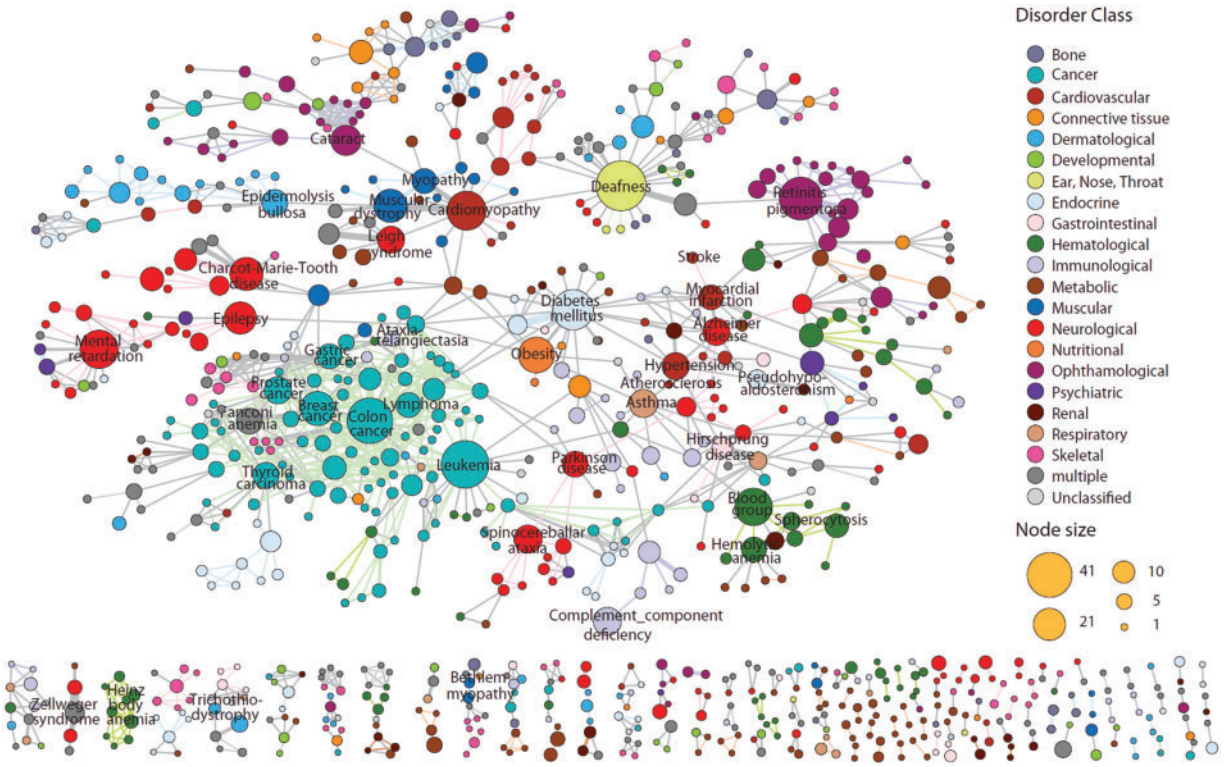


Figure 2: Graph representation of the human disease network. Two disease nodes (circles) are connected if they share a common genetic component according to disorder disease-gene associations listed in OMIM as of the year 2005. Adapted from [14]. Copyright (2007) National Academy of Sciences, USA.

hypothesis models a disorder as the result of perturbations or breakdown of a specific functional module caused by variations in one or more of the components (rather than that of a singular component). It can provide a network-based interpretation and understanding of complex diseases, but precise definition and identification of disease modules still remain to be established.

The modular nature of disease can be exploited in predicting new disease genes. According to the network-based modular model of disease, genes that are functionally related with known disease genes for a specific disorder should have higher odds to be involved in the same functional module and thereby the same disease, providing a way of prioritizing disease gene candidates. A proof of principle example of this scenario came from the complement component *C5*. As of 2005, it was not designated as a disease gene for complementary deficiency disorder, but was reported to interact with five other complement components associated with the disease. After further evidence, it was finally listed as a validated disease gene for the disease in the 2009 version of OMIM. Similar ideas have been implemented to various disease gene prediction algorithms, including the methods based on the proximity of proteins in the protein networks, and integration of large-scale genomic and phenotypic data, which are reviewed in a recent review article [38]. The network-based idea has also been combined with other omics techniques for disease gene identification. For example, mapping protein–protein interactions starting from known disease genes for specific disorders has proved useful in identifying potential disease factors in Huntington's disease [39], ataxia [40] and breast cancer [41]. Combining gene expression profiles from disease samples to construct the underlying gene network was used to unveil the genetic mediator in the prostate cancer progression [42]. It was shown that most known biomarkers for metabolic diseases can be justified by metabolic flux analysis of the reconstructed human metabolic network, opening a possibility for using metabolic network to disease-gene discovery [43]. All these examples illustrate a promise for the disease module-based network approach to provide a useful tool in disease-gene identifications, as well as in realizing the network therapy or the 'network

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drug', a network-based therapeutic strategy that targets the disease module as a whole rather the individual component within it [44].

Highly connected central proteins (so-called 'hub' proteins) in the interactome network have been thought to play essential roles in basic functionality of the cell, from the studies on model organisms such as budding yeast [45]. Given the potentially severe phenotypes arising from mutations in human disease genes, one might assume the same centrality–essentiality relationship for the disease genes [46]. From a careful analysis by distinguishing human essential genes from human disease genes, a more complicated picture emerged for human [14, 22, 47]. Essential genes are indeed found to more likely encode hub proteins and are widely expressed, agreeing with results from model organisms. Once one controls the fact that some disease genes are also essential genes, the non-essential disease genes display a very different trend in that they do not tend to encode hub proteins and are specifically expressed (Figure 3). These observations suggest that human disease gene products are functionally and topologically peripheral in the cellular network, being less likely to participate in vital cellular functions, and are dynamically less connected to the rest of the cellular system. In contrast, essential genes appear to play a topologically and functionally central role, with an expression pattern that is highly integrated with the expression of other genes. Distinct characteristics of human essential and disease genes at the genomic level have also been documented [48–50].

The origin of such peripherality of human disease genes was understood by the following selection-based argument [14]. Disorders covered in the OMIM database are dominantly genetic, inherited disorders. Therefore, for a gene to be an OMIM disease gene, the mutation(s) for the disease has to be inherited, requiring that the carriers of that mutation should live at least up to his/her reproductive age. Those mutations giving rise to developmental impairments severe enough to lead to *in utero* lethality—lethal mutations of essential genes—are hard to survive and therefore unlikely to be inherited. Therefore, OMIM disease-related mutations are more likely to reside in the functionally and topologically peripheral regions of the cell, giving a higher chance of viability. This selection-based argument for the depletion of central inherited disease genes implies the counter-argument for disease mutations that are less subject to selective pressure. Not surprisingly, the same centrality–essentiality analysis of the currently known set of somatic cancer genes reveals that indeed they are highly central, although the general conclusion requires more data [51]. Further graph-theoretical analyses including other centrality measures such as closeness and coreness with disease genes identified from genome-wide association studies [28] and on GAD disease genes (unpublished) also conform to this argument.

EXTENSIONS OF THE HUMAN DISEASE NETWORK

A natural extension of the diseasome idea is to include drugs. To this end, the so-called drugome has been constructed, as the combined set of available drug chemicals and their target genes, by a bipartite network connecting them [25]. Overlaying the drugome on the diseasome through common genetic components, some interesting observations were made. First, currently available drugs do not target diseases equally, but are found to be disproportionately enriched in some regions of the diseasome, confirming prevalence of 'me-too' drugs on the market. Second, although the enrichment for etiological drugs which directly target the disease-causing component was clearly observed, still a majority of existing drugs target components as far away from the disease-causing genes as a random target would do, suggesting a predominance of palliative-acting drugs. Third, there is a significant shift towards the closer-to-target drugs approved after 1996, compared with those approved before 1996, supporting a move towards rational drug design. Aside from these interesting findings, the drugome information would be an essential additional layer to the diseasome, with which we can assess the current status of understanding and treatment of human diseases at a global scale, guiding future roadmaps for drug discovery [52].

In the original diseasome map (Figure 2), metabolic disorders do not form a single distinct cluster but represent the least connected disorder class. They are under-represented in the largest cluster but over-represented in the small clusters. It was, however, later shown that the metabolic disorders are more connected metabolically through adjacent metabolic pathways, rather than genetically through sharing common disease gene mutations [26]. With the so-called metabolic disease network (Figure 4A), in which two diseases are connected if the metabolic

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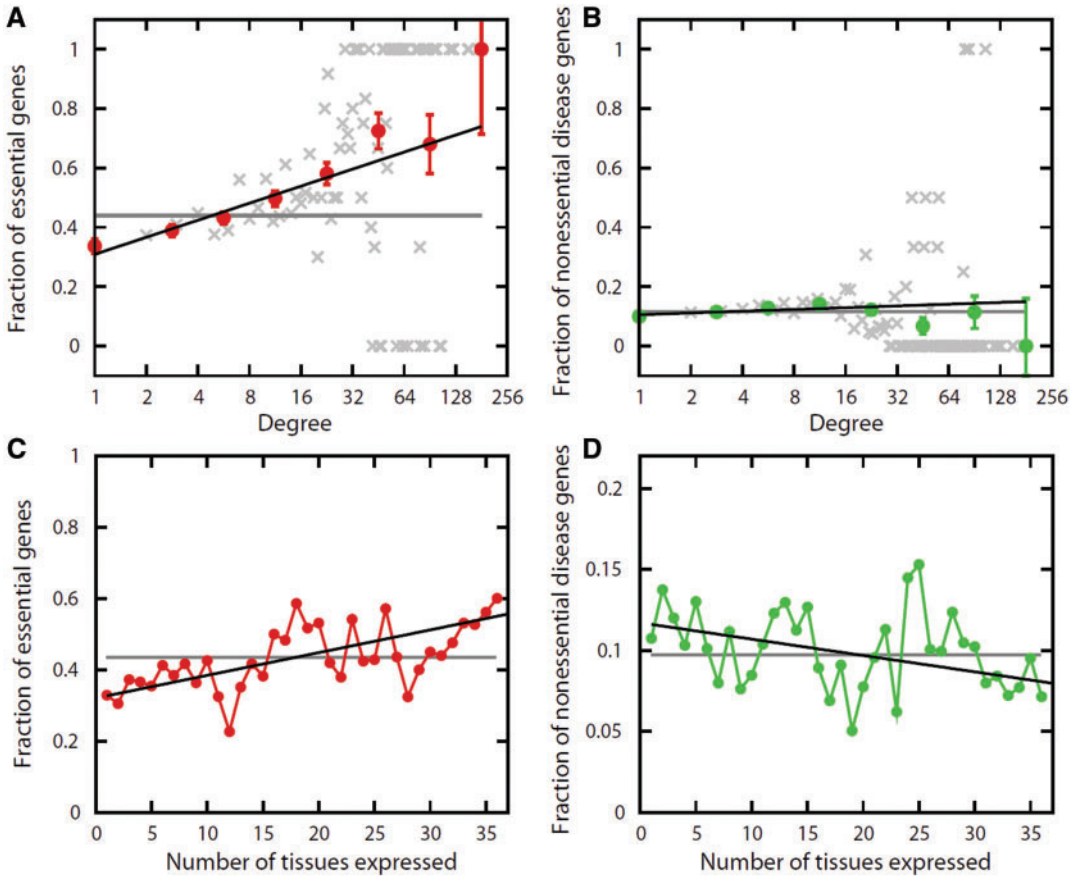


Figure 3: Centrality–essentiality analysis of human genes. Fraction of essential genes increases with the degree (number of links) in the protein–protein interaction network, implying that the human essential genes are more likely to encode hub proteins (**A**). When controlled for the essentiality, the non-essential disease genes do not show the trend (**B**). Also, essential genes are enriched for widely expressed genes (**C**), whereas non-essential disease genes are more likely specifically expressed across human tissues (**D**). Adapted from [14]. Copyright (2007) National Academy of Sciences, USA.

reactions they are associated with are adjacent [26], it is also shown that two diseases connected by metabolic links exhibit higher comorbidity, that is, they are more likely to occur in the same patient than expected at random, suggesting a shared pathophysiology between those diseases. This is one example of a more general scenario that components and factors other than purely genetic ones can also play a key role in some disease processes. Non-genetic biomolecules such as steroids play an important role in many metabolism-related processes in the body [53]. The relevance of exogenous non-genetic factors such as the environmental factors and non-biological substances is being increasingly documented [54]. Even in the genetic part, the Mendelian component constitutes only one, albeit significant, portion. Effect of somatic mutations is being increasingly documented, notably for cancers [51]. A recent surge of genome-wide association studies attempt to identify multi-component genetic factors for complex diseases that are truly polygenic and thus cannot be attributed to an individual Mendelian factor. There are also phenotype-driven approaches. For example, the so-called human phenotypic disease network can be constructed (Figure 4B), in which two diseases are connected by the so-called comorbidity link when the two diseases occurred in the same patient with higher frequency than expected at random [27]. It is found that diseases with more comorbidity links are more likely to be lethal (the complete comorbidity dataset used in the study has been made available at http://hudine.neu.edu). Various other concepts were introduced in the context of disease and networks [55–57]. Interaction between human and viral proteins can also be integrated for understanding of viral diseases [58, 59]. Eventually, to build an

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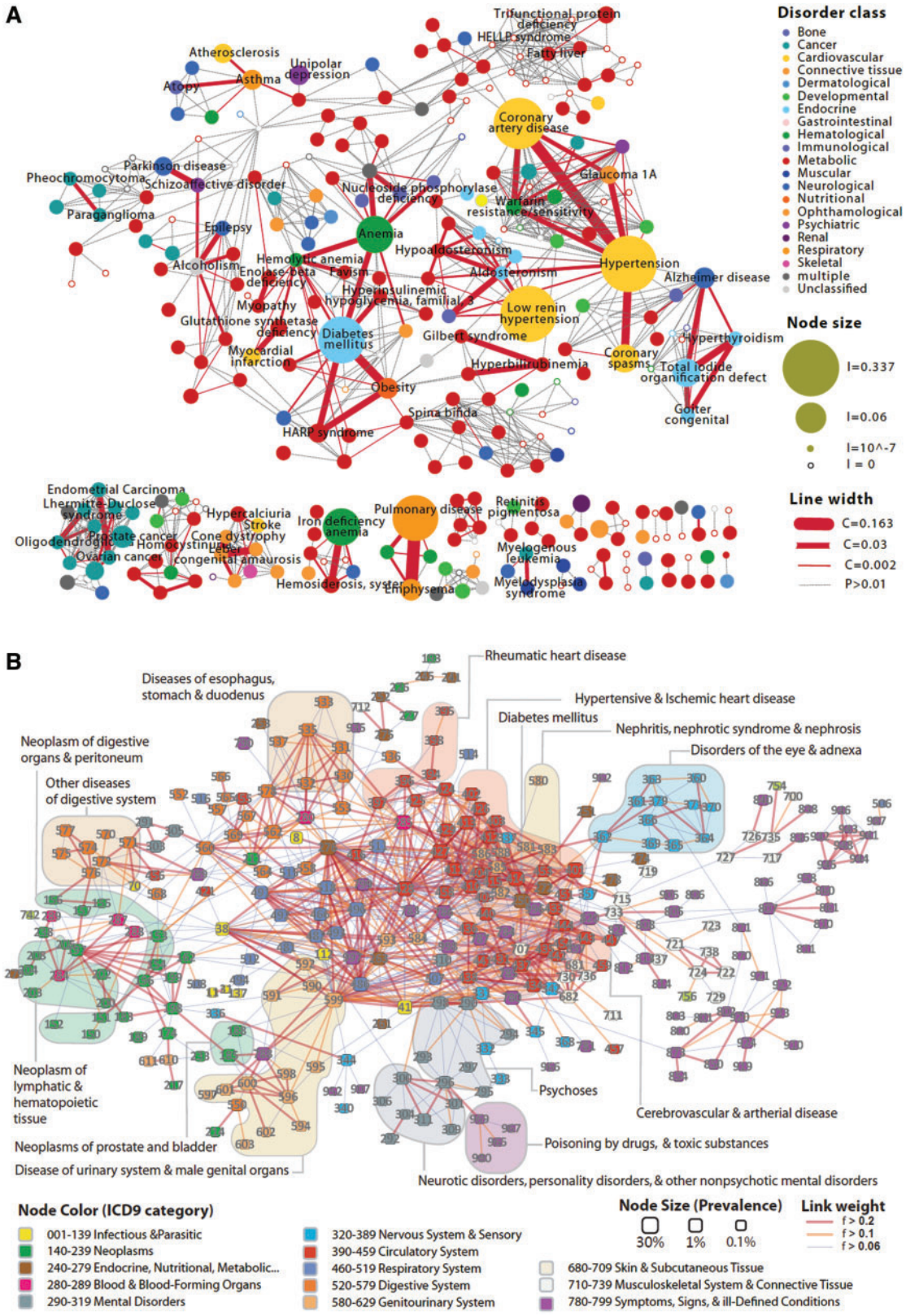


Figure 4: Graph representation of the human metabolic disease network (**A**) and the human phenotypic disease network (**B**). In (A), two diseases associated with adjacent metabolic reactions are connected by a link, the width of which is coded by the comorbidity score. In (B), two diseases are connected if they have comorbidity score (via ϕ-correlation) higher than the preset threshold value. Adapted from [26] and [27], respectively. Copyrights (2008) National Academy of Sciences, USA, and (2009) Public Library of Science, respectively.

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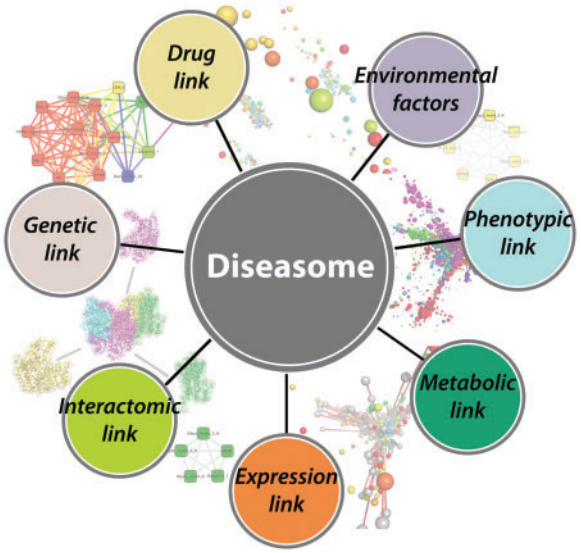


Figure 5: A roadmap towards the complete diseasome. Each and every disease-contributing factor such as molecular links from interactome, co-expression and metabolism, as well as genetic interactions and phenotypic comorbidity links, will have to be integrated in a context-dependent manner. Furthermore, drug chemical information and non-biological environmental factors such as toxicity information altogether must also be incorporated.

ultimate map of human diseases, all these factors, and presumably more to come, should eventually be incorporated, which is the big challenge ahead.

CONCLUSIONS AND FUTURE PROSPECTS

Like many other branches of science in the 20th century, the reductionism paradigm has dominated biology and medicine. Rather, the network approach to human diseases aims to view human diseases as a whole. This complementary new approach offers the possibility of discerning general patterns and principles of human disease not readily apparent from the study of individual disorders. A useful tool in this approach is the concept of the human diseasome that represents a genome-wide relationship between known diseases and associated genetic components. It can provide a roadmap for future studies on disease associations not only for basic biomedical researchers but also for clinical practitioners. It is obvious, however, that any current version of the human diseasome would be by no means complete. New development of technology and analytical tools will constantly produce new information on human variations that will add new links to the network. This may change the detailed picture of the human diseasome organization that currently exists; it may push some currently isolated disorders into larger clusters and even new nodes (disorders and disease genes) will appear on the map over time. Nonetheless, its key characteristics such as modular diseasome hypothesis and peripheral nature of disease genes are expected to be robust.

The network concept of human disease is an inevitable extension of the network concept of fundamental cell biology. It is already starting to stimulate transitions in how we view human genetics, how we approach treatment of genetic disorders and how we formulate and integrate models of human systems biology. As reviewed here, more and newer data on polygenic associations on complex disorders, non-genetic biological diseases-causing components and exogenous components such as environmental factors are constantly becoming available. Combined with all such information, the human diseasome and human disease network will get closer to a more complete systemic atlas of human diseases (Figure 5). In so doing, several conceptual and computational hurdles will have to be overcome. For example, we have to formulate the context-dependent model of integrated multiplex interactome networks and the integrative multi-relational diseasome network. Definition and identification of disease module is also a pivotal step for which we currently have only an incomplete answer. Collective efforts from diverse disciplines will be the key to the success of this program. We hope this review article brings wider attention from interested researchers to realize the full potential of the new discipline of 'network medicine,' including a fully rational and principled categorization of human diseases [60].

Key Points

- Human diseases can be 'connected' through various molecular and genetic associations.
- Network approach to human diseases as a whole—the human diseasome—is proposed.
- Human diseases are hypothesized as a disruption of 'disease module' in the interactome network.
- Essential and disease genes occupy distinct functional locations in the interactome network.
- Integration of various disease-associated biological and non-biological factors is necessary to build the complete human diseasome.

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